

Shubham Sinha

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Experience

Numerator

Jan 2024 – Present

Engineer 1

Remote, India

- Spearheaded the collaborative agile development of Business Intelligence platforms, enabling data associates to seamlessly explore, interact, and modify core data. Achieved a significant **50%** increase in overall performance compared to previous solutions, optimizing query time for a billion-row scale database from minutes to seconds.
- Engineered a scalable **Python** based service-oriented architecture using **Django** and **Elasticsearch**, decreasing API request latency by **150ms** and handling **30** concurrent users with zero downtime.
- Established a robust change data capture (CDC) system between MySQL and ES with the Publish-subscribe pattern and **Celery** to optimize data processing.
- Leveraged **transformers** from the **Hugging Face** library for **fine-tuning** pre-trained models and **spaCy** for text pre-processing to predict the brand and category of the newly ingested US, CA and German products that increased the item attribution rate by **23%**.

Inexture Solutions LLP

Dec 2020 – Dec 2023

Python Developer

Ahmedabad, Gujarat, India

- Worked with India's leading cryptocurrency app with 19 million registered users. Built **UPI** payment integration enabling users to deposit funds into their app wallets that reduced downtime by **25%** and improved customer experience.
- Created real-time monitoring dashboards with **Grafana** and **ELK**, enabling rapid detection of anomalies and decreased the average time to resolution (MTTR) by **40%** which led to improved payment processing speeds.
- Developed Producer and Consumer using **RabbitMQ** to transfer data between **microservices** for India's leading online employment solution application.
- Executed **Jenkins CI/CD** pipelines, automating multi-stage builds and deployments, eliminating manual intervention by **80%** and accelerated feature delivery.
- Automated GitHub-to-Jenkins workflows with real-time build notifications, ensuring faster feedback loops and lowering code-to-production cycle time.

Technical Skills

Languages: Python, C, C++, Java, Go, JavaScript

Frameworks: Django, Django CMS, Flask, Django REST Framework, FastAPI

Artificial Intelligence: Deep Learning, Large Language Models (LLM), RAG, LangChain, Fine-tuning, Prompt Engineering

Tools: Docker, Kubernetes, Git, Kibana, New Relic, Cloudinary, Logtail, RabbitMQ, Elasticsearch, Airflow

Database: MySQL, PostgreSQL, SQLite, MongoDB, Chroma, Pinecone, Faiss

AWS Services: Elastic Beanstalk, CodePipeline, EC2, SES, CloudWatch, Lambda

Projects

AI Trading Recommendation Engine | *Python, CrewAI, Groq LLM, Streamlit, yfinance, pandas, regex*

- Developed a multi-agent AI stock analysis system using the **CrewAI** framework with specialized financial analyst and trader agents powered by Groq's **Llama-3.1-8b-Instruct** LLM.
- Constructed interactive **Streamlit** web application providing real-time stock metrics, trading recommendations, and historical visualizations. Integrated Yahoo Finance API for live market data and **Groq** AI for intelligent analysis with zero-temperature deterministic outputs.

Medical Knowledge Retrieval System | *Python, Flask, LangChain, Groq, Pinecone, PyPDF*

- Engineered intelligent medical **chatbot** using **RAG** architecture with **Flask**, **Pinecone** vector database and HuggingFace embeddings. Incorporated Groq's **Llama 3** model via **LangChain** for contextual medical responses.
- Processed PDF documents into searchable knowledge base with end-to-end pipeline delivering accurate medical responses through web interface.
- Implemented **CI/CD** pipeline with **GitHub Actions** for automated testing, linting, and security scanning.

Education

Dharmsinh Desai University

Bachelor of Technology in Information Technology (IT)

Nadiad, Gujarat, India

August 2017 – May 2021